

Flying Liferaft

UAV propeller analysis and actuator sizing.



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Abstract

Every year there are over two thousand world-wide drowning deaths at sea, 55% of which are migrants trying to cross the Mediterranean. An unmanned aerial vehicle (UAV) for delivery of flotation aid within 30 minutes is proposed as a novel and effective tool for coast guard operators to save lives. The drone has a 51.5 kg maximum take-off weight (MTOW), hybrid-electric propulsion, dual fixed-wing and quad-copter configuration and is capable of vertical take-off and landing (VTOL). This report explores the initial mass estimation of the UAV and creation of a constraint diagram to obtain thrust and wing loading requirement. The propeller performance is analyzed to ensure successful operation at all flight regimes in the mission profile and further optimizations to the blades are described based on varying Reynolds number. Additionally, a finite element analysis of the blades is carried out to ensure structural integrity and optimum performance by validating analytical results and quantifying aeroelastic effects. Finally, the flight control surface actuators are selected and integration mechanisms are sized to satisfy control requirements while minimizing mass and cost. In summary, analysis of the propulsive system of the UAV shows very promising results for potential rapid delivery of aid in any maritime environment at all weather conditions.

Contents

List of Figures	iv
List of Tables	iv
List of Symbols	v
1 Introduction	1
2 Initial Sizing	1
2.1 Weight Estimation	1
2.2 Forward Flight Constraints	1
3 Propeller Governing Theory	2
3.1 Blade Element Momentum Theory	2
3.2 Tip Losses	3
3.3 Varying Reynolds Number	3
3.4 Post-stall Performance	4
4 Propeller Performance	5
4.1 Pusher Thrust Lapse	6
4.2 Effect of Forward Velocity on VTOL Propellers	6
4.3 VTOL Thrust Lapse	7
4.4 VTOL OEI Condition	8
5 Finite Element Analysis	8
6 Control Surface Actuators	10
7 Discussion and Further Steps	11
References	12
Appendix	13
A VTOL Thrust Lapse With Altitude	14
B Actuator Drawings	14

List of Figures

1	Baseline configuration CAD [5].	2
2	Forward flight constraint diagram.	2
3	Velocities through a rotor disk annulus.	2
4	Perceived airflow at blade section.	2
5	Pusher blade CAD.	3
6	VTOL blade CAD.	3
7	Variation of Reynolds number along blades.	3
8	Lift to drag ratio along pusher blade.	4
9	Lift to drag ratio along VTOL blade.	4
10	Variation of lift coefficient with angle of attack of E216 airfoil at $Re = 100000$	5
11	Variation of drag coefficient with angle of attack of E216 airfoil at $Re = 100000$	5
12	Variation of conditions along pusher blade.	5
13	Variation of conditions along VTOL blade.	5
14	Variation of maximum thrust with freestream velocity for pusher propeller at $h = 0$	6
15	Variation of maximum thrust with altitude for pusher propeller at $V_\infty = 58.3$ m/s.	6
16	Variation of thrust on a VTOL blade as it spins at $V_{forward} = 37$ m/s.	7
17	Variation of total thrust on a VTOL propeller as it spins at $V_{forward} = 37$ m/s.	7
18	Variation of VTOL maximum thrust with changing V_{climb} at fixed power.	7
19	Variation of VTOL maximum thrust with changing $V_{forward}$ at fixed RPM.	7
20	Von Mises stress on pusher blade in FEA.	9
21	Von Mises stress on VTOL blade in FEA.	9
22	Side view of elevator actuator system. All sizes in mm.	11
23	Top view of elevator actuator system. All sizes in mm.	11
24	Variation of VTOL maximum thrust with altitude.	14
25	Side view of elevator actuator. All sizes in mm.	14
26	Side view of rudder actuator. All sizes in mm.	14

List of Tables

1	Actuator properties.	11
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List of Symbols

a	Axial Induction Factor
a'	Tangential Induction Factor
α	Angle of Attack
$\mathcal{AR}$	Aspect Ratio
B	Number of Blades
β	Blade Twist Angle
c	Chord Length
C_d	Drag Coefficient
C_{d_0}	Parasitic Drag Coefficient
C_{d_1}	Pre-stall Drag Coefficient
C_{d_2}	Post-stall Drag Coefficient
C_l	Lift coefficient
C_{l_1}	Pre-stall Lift Coefficient
C_{l_2}	Post-stall Lift Coefficient
D	Drag
e	Oswald Efficiency
E	Young's Modulus
F	Tip-Loss Factor
ϕ	Inflow Angle
g	Gravitational Acceleration
h	Altitude
I	Second Moment of Area
L	Lift
L_{horn}	Control Horn Length
L_{Arm}	Servo Arm Length
n	Load Factor
P	Power
P_0	Power at Sea Level
P_{crit}	Buckling Load
Q	Torque
r	Radial Position
Re	Reynolds Number
ρ	Density
ρ_0	Density at Sea Level
S_{ref}	Density
t	Time
T	Thrust
T_{act}	Actuator Moment
T_{hinge}	Hinge Moment
V	Perceived Velocity at Blade
V_i	Induced Velocity
V_∞	Freestream Velocity
W	Weight
W_0	Maximum Takeoff Weight
W_e	Empty Weight Fraction
W_f	Fuel Weight Fraction
W_p	Payload Weight Fraction
ω	Rotational Velocity

1 Introduction

This report explores the design and analysis of a fixed wing/VTOL hybrid-electric UAV capable of providing flotation aid at sea. After different trade-off analyses, the mission profile was selected to be a 70 km cruise at 210 km/h, a loiter time of 10 min and a second cruise also of 70 km. The drone operates from a rescue ship and must be capable of vertical takeoff and landing. Therefore, the baseline configuration was selected to be four vertical propellers for VTOL and one pusher for forward flight, as well as fixed wings and an H-tail. The baseline configuration can be seen in Figure 1. The design payload carrying capacity was set at 9 kg to be able to carry individual buoys or a large life raft.

Within the Propulsion and Systems subteam, this report details the individual work carried out consisting of the initial weight estimation, the initial sizing for the fixed wing mode of the aircraft, the aerodynamic analysis and optimization of the propellers, the finite element analysis of the propeller blades and the control surface actuator sizing and integration.

2 Initial Sizing

2.1 Weight Estimation

The initial stage in aircraft design is the initial weight estimation which was performed using regression data from previous similar vehicles. Firstly, the empty weight fraction can be estimated as $W_e/W_0 = AW_0^C$ [1] where $A = 0.7$ and $C = -0.06$ for UAVs of similar size [2]. The fuel fraction was estimated using the Breguet range and endurance equations [1], which yielded an initial fuel mass estimate of 1.96 kg. Additionally, the battery weight fraction was estimated to be 21% as detailed in [3].

Having appropriate expressions for the weight fractions and a fixed payload weight of 9 kg for the mission, the MTOW of the UAV was determined to be 51.5 kg. Including an empty weight fraction of 78.7% (including batteries), a payload weight fraction of 17.5% and a fuel fraction of 3.8%. The empty weight fraction is large compared to most aerospace vehicles due to the dual fixed-wing and VTOL configuration, especially so because of the batteries. However, the fuel fraction is small compared to most vehicles as the required range is very low in proportion to the required performance and payload capacity. These initial weight estimations were used by the different subteams to start designing the UAV. As the design process progressed, more accurate weight estimations were made from the actual components of the aircraft as explained in weight and balance in [4].

2.2 Forward Flight Constraints

After determining an initial estimate of the mass, it is imperative to make an initial estimation of the required propulsive and aerodynamic capabilities of the aircraft to satisfy the necessary performance metrics, these estimates were then used as the target design by the different subteams, especially Aerodynamics, Structures, and Propulsion. The target values can later be refined when more accurate values for aerodynamic properties of the drone are available, in particular parasitic drag and aspect ratio.

From force balance, Equation 1 gives the required thrust for a given flight condition or manoeuvre with fixed aerodynamic properties, all of which were estimated from historical data [1][2].

$$\frac{T}{W} = \left[\frac{1}{V_\infty} \frac{dh}{dt} + \frac{1}{g} \frac{dV_\infty}{dt} + \frac{0.5\rho V_\infty^2 C_{d0}}{\alpha W_0/S_{ref}} + \frac{\alpha n^2 W_0/S_{ref}}{0.5\rho V_\infty^2 \pi Re} \right] \quad (1)$$

The required thrust for a given manoeuvre is given by Equation 1 by setting the climb rate $\frac{dh}{dt}$, acceleration rate $\frac{dV_\infty}{dt}$ and the load factor n . The necessary flight capabilities are the first and second cruise, loiter, and a climb capability of 500 ft/min at the cruise speed. The climb requirement was set to be able to respond quickly to gusts as the drone must be operable in all weather conditions. This also allows fast climb manoeuvres during transition. Additionally, a maximum stall speed of 29.7 m/s was set by the Aerodynamics subteam for safety during loiter based of wind gust data. This also allows to reduce the transition time and therefore the battery size. Figure 2 shows the constraint diagram, including all constraint curves and the chosen design point.

Based on the constraints, the required thrust to weight ratio of the UAV is 0.22 (at $V_\infty = 58.3$ m/s) and the wing loading is 1006 N/m². This gave the design pusher propeller requirement of 111 N thrust at cruise conditions. Even though the required thrust at other conditions such as loiter is lower, it is expected that the available thrust at this lower speed will be less than at the cruise condition as the propeller is optimized for cruise speed. Therefore, after designing the propeller, it must be ensured that its off-design performance is sufficient for the other constraints. The initial sizing and requirements for the VTOL propellers were done separately and are explained in [4], giving a minimum required thrust of 152 N per propeller.



Figure 1: Baseline configuration CAD [5].

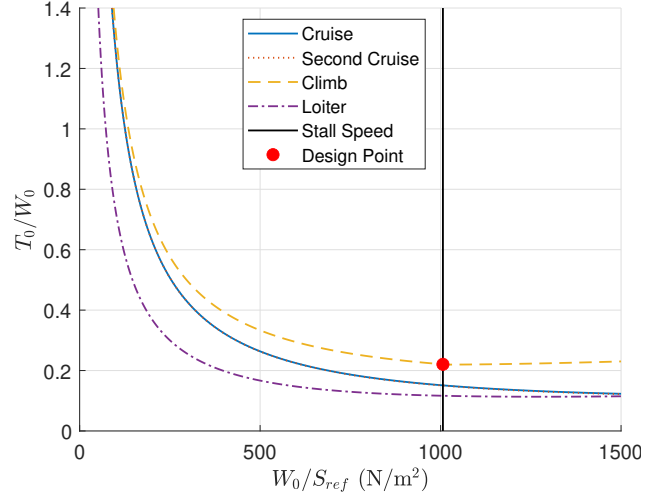


Figure 2: Forward flight constraint diagram.

3 Propeller Governing Theory

As shown in Figure 1, the drone has five propellers. The first is a pusher propeller which is located behind the horizontal stabiliser and is used during forward flight and is powered by an internal combustion engine. Additionally, the aircraft has four VTOL propellers oriented vertically for vertical takeoff, landing, and hover, these are powered by four electric motors.

A blade element momentum code was developed to predict the performance of fixed geometry propellers at given conditions. Its purpose was to test the performance of the propeller at off-design environments, as well as calculating the force and moment distribution for following structural considerations. This is in contrast to the optimization performed by M. Hamburger in [6] which optimizes the blade geometry for a given thrust requirement.

3.1 Blade Element Momentum Theory

Blade element momentum theory equates the momentum change of the fluid passing through an annulus of the propeller disk to the force produced by the section of the propeller blades which passes through said annulus [7]. Figures 3 and 4 illustrate the flow through an annulus and the perceived airflow at a blade section.

Momentum theory shows that the induced velocity at the wake is twice the induced velocity at the rotor [8].

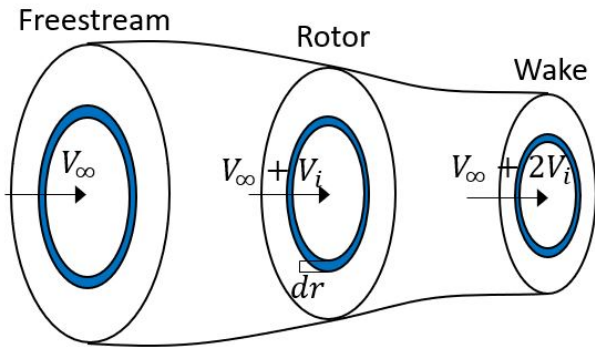


Figure 3: Velocities through a rotor disk annulus.

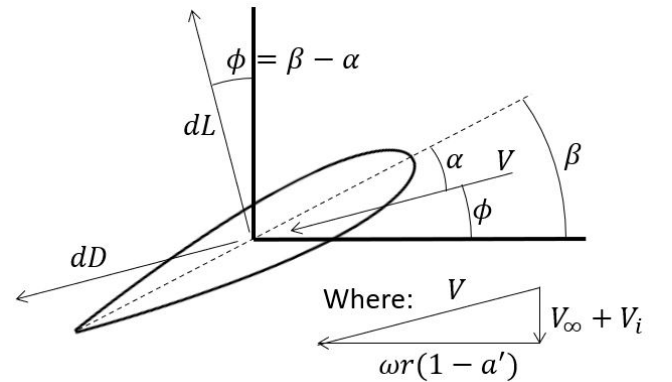


Figure 4: Perceived airflow at blade section.

Overall, by equating the thrust and torque at each blade section to the momentum change in its annulus, BEM theory yields the relations in Equations 2 and 3.

$$dT = 4\pi r \rho V_i(r)(V_\infty + V_i(r))Fdr = \frac{B\rho c(r)V^2}{2} (C_l(\alpha, Re) \cdot \cos(\phi) - C_d(\alpha, Re) \cdot \sin(\phi)) dr \quad (2)$$

$$dQ = 4\pi r^3 \rho (V_\infty + V_i(r)) a' \omega F dr = \frac{B \rho c(r) V^2}{2} (C_l(\alpha, Re) \cdot \cos(\phi) - C_d(\alpha, Re) \cdot \sin(\phi)) dr \quad (3)$$

Addition of the different velocity components gives the flow conditions at each blade section as shown in Equation 4.

$$V = \sqrt{(V_\infty + V_i)^2 + (\omega r(1 - a'))^2} \quad \text{and} \quad \phi = \tan^{-1} \left(\frac{V_\infty + V_i}{\omega r(1 - a')} \right) \quad (4)$$

Starting with an initial guess for induced velocity V_i and induced tangential velocity factor a' allows calculation of conditions at each blade section with Equation 4 and therefore lift and torque generated which in turn allow recalculation of the induced velocities using Equations 2 and 3. Iterating until convergence yields the correct aerodynamics conditions along the blade.

Blade element momentum theory has significant assumptions, on its own it does not consider the tip and hub effects. However, the tip loss factor F is introduced to account for tip effects. It also assumes that the flow in each annulus is independent of the other annuli, this is not true due to the viscosity and pressure gradient in the fluid, however, these effects are generally negligible as the induced velocity is optimized to be approximately constant along the blade. A further assumption is that all flow is laminar, this assumption should hold, even in the pusher behind the fuselage, as the propellers generate favourable pressure gradients.

3.2 Tip Losses

The tip loss factor F was introduced in Equations 2 and 3 to account for tip-loss effects. Blade element theory assumes the flow on each blade element to be independent to the neighbouring sections. This becomes very inaccurate close to the tip, as having a finite blade implies that the pressure on both sides of the blade must be equal at the tip. The analytical expression for F , as calculated by Prandtl, is shown in Equation 5 [9].

$$F = \frac{2}{\pi} \cos^{-1} \left(\exp \left(-\frac{B}{2} \left(1 - \frac{r}{R} \right) \sqrt{1 + \frac{V_t^2(R)}{V_n^2(R)}} \right) \right) \quad (5)$$

Where V_t and V_n are the tangential and axial velocity respectively at the tip of the blade. Several approximations exist for the tip-loss factor such as the ones suggested by Prandtl, Glauert or Burton [9]. The numerical code, however, uses the exact solution as the post-stall characteristics (Section 3.4) allow convergence of the code at all conditions. The tip-loss factor along the blades will be shown in Figures 12 and 13.

3.3 Varying Reynolds Number

An important consideration for propeller design is the variation of the operating conditions along the blade. The change in Reynolds number along the blades is shown in Figure 7. It can be seen that the Reynolds number varies from about 50000 to 230000 for the pusher and from 50000 to 330000 for the VTOL blades. This causes strong changes in the lift and drag coefficients and to the maximum lift to drag ratio.

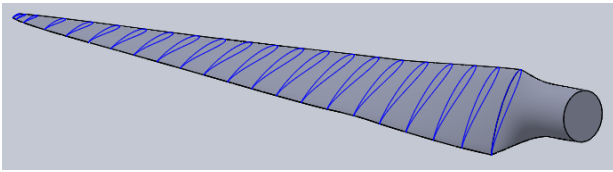


Figure 5: Pusher blade CAD.

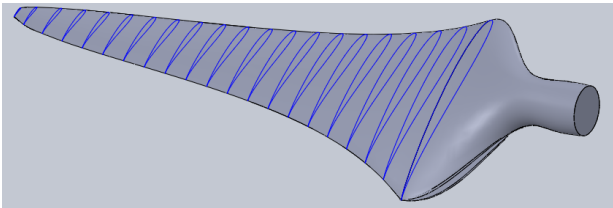


Figure 6: VTOL blade CAD.

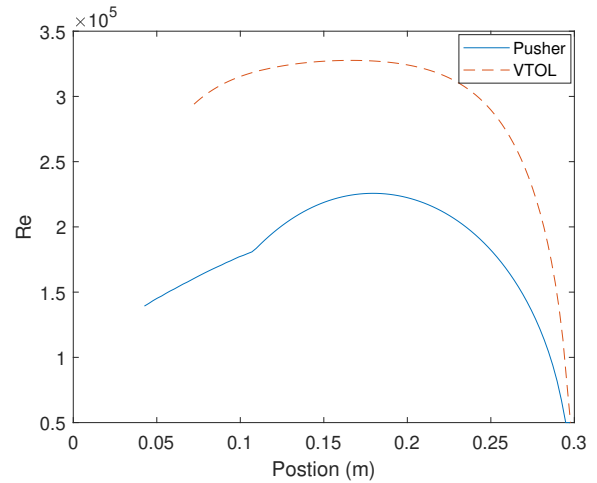


Figure 7: Variation of Reynolds number along blades.

The code for blade optimization from Adkins [10] used in [6] assumes a constant Reynolds number along the blade and therefore the design is suboptimal. The polars of the E216 airfoil were computed for a range of Reynolds numbers within the operating conditions using XFOIL [11] viscous simulation up to stall angles of attack.

Using the computed data, the twist distribution was updated to increase the L/D ratio. The change in conditions after this optimization can be seen in Figures 8 and 9. This led to a decrease in required power of 1.1% for the pusher propeller and 14.1% for the VTOL propellers at design conditions. The change is larger for the VTOL propeller because it has a wider Re range. The improvement is also more consequential VTOL power requirements cause a very large increase in aircraft weight.

The reasons for the small mismatch which remains between L/D and $(L/D)_{max}$ include the accuracy of the fits and the shape modifications due to structural considerations [3]. This has its largest effect near the root of the pusher blade.

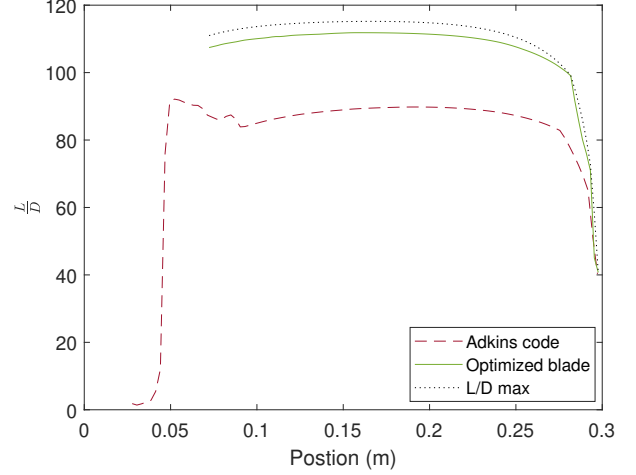
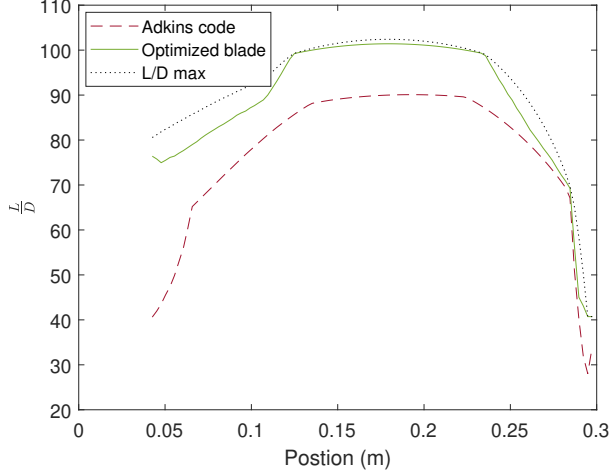


Figure 8: Lift to drag ratio along pusher blade. Figure 9: Lift to drag ratio along VTOL blade.

3.4 Post-stall Performance

The simulations in XFOIL to generate the lift and drag curves can only be performed when the flow is attached to the airfoil. Therefore they provide no data after stall or at large negative angles of attack. The post-stall performance is required to obtain reasonable values for torque and thrust at off-design conditions, especially at lower RPMs where a significant portion of the propeller is stalled.

A commonly used idealization is to treat the airfoil as a flat plate after stall. However, it has been shown that this approximation underestimates both lift and drag at large angles of attack [12]. As a better approximation, the AERODAS [12][13] model was used to calculate the airfoil aerodynamics at post-stall angles. The expressions for lift and drag after stall predicted by the model are shown in Equations 6 and 7.

$$C_{l_2} \approx -0.032(\alpha - 92) - (1.632 - C_{l_2max}) \left(\frac{92 - \alpha}{51} \right)^{N_2} \quad \text{where } N_2 = 1 + \frac{C_{l_2max}}{1.632 - C_{l_2max}} \quad (6)$$

$$C_{d_2} \approx C_{d_1max} + (C_{d_2max} - C_{d_1max}) \sin \left(90 \cdot \frac{\alpha - \alpha_{maxD_1}}{90 - \alpha_{maxD_1}} \right) \quad (7)$$

Where all angles are in degrees, C_{l_1max} and C_{d_1max} are the maximum coefficients before stall and α_{maxD_1} is the angle of attack at which maximum pre-stall drag occurs [12]. The previous pre-stall parameters are obtained through the XFOIL simulation at each Reynolds number. C_{l_2max} and C_{d_2max} are the maximum post-stall lift and drag coefficients for which the AERODAS models provides the expressions shown in Equation 8 [13].

$$C_{l_2max} = 1.19 \left(1 - \left(\frac{t}{c} \right)_{max}^2 \right) \quad \text{and} \quad C_{d_2max} = 2.3 \exp \left[- \left(0.65 \left(\frac{t}{c} \right)_{max} \right)^{0.9} \right] \quad (8)$$

The post-stall maximum coefficients from Equation 8 depend only on the maximum thickness to chord ratio $(t/c)_{max}$ and do not account for camber. This is one of the main limitations of AERODAS. However, it is much more accurate than a flat-plate representation. The results close to design conditions need to be very accurate

to get correct performance values for the overall aircraft. Since AERODAS's approximations are not used at pre-stall conditions, the results are not compromised.

The extended lift and drag curves for the E216 airfoil, the chosen airfoil for the propellers, can be seen in Figures 10 and 11.

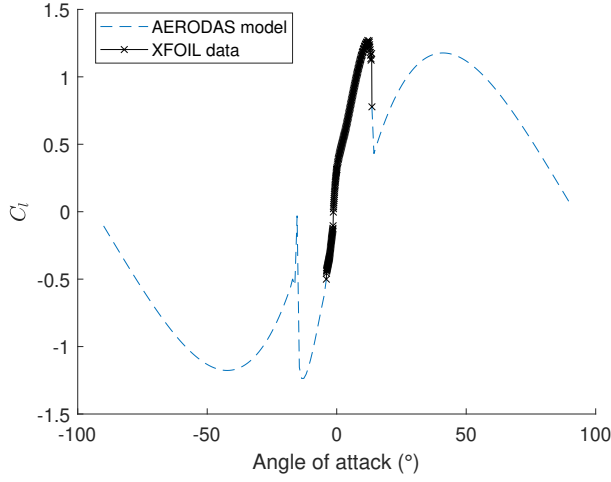


Figure 10: Variation of lift coefficient with angle of attack of E216 airfoil at $Re = 100000$.

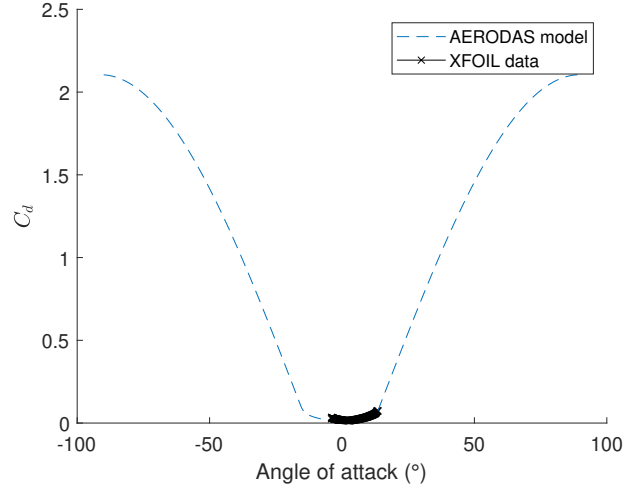


Figure 11: Variation of drag coefficient with angle of attack of E216 airfoil at $Re = 100000$.

4 Propeller Performance

The induced velocity factor $a = V_i/V_\infty$, tangential induction factor a' , and tip-loss factor F along a pusher blade at the design cruise velocity of 58.3 m/s and design RPM of 6500 can be seen in Figure 12. The same parameters for the VTOL motor at hover with an RPM of 6000 are shown in Figure 13. However, the induced velocity is shown instead of the induction factor as the freestream velocity is zero during hover. The mean induced velocity along the rotors was required by the Control subteam to analyze interaction with the wings and horizontal tailplane. The mean induced velocity by the rotor disc was calculated as a weighted average of the induced velocity where the weights scale proportionally to the radial position as the area of each annulus is directly proportional to the radial position. The mean induced velocity was found to be 3.98 m/s for the pusher propeller and 16.0 m/s for the VTOL propeller at design conditions.

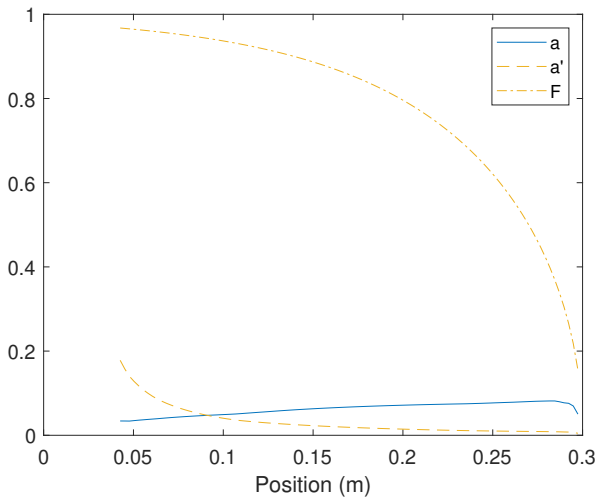


Figure 12: Variation of conditions along pusher blade.

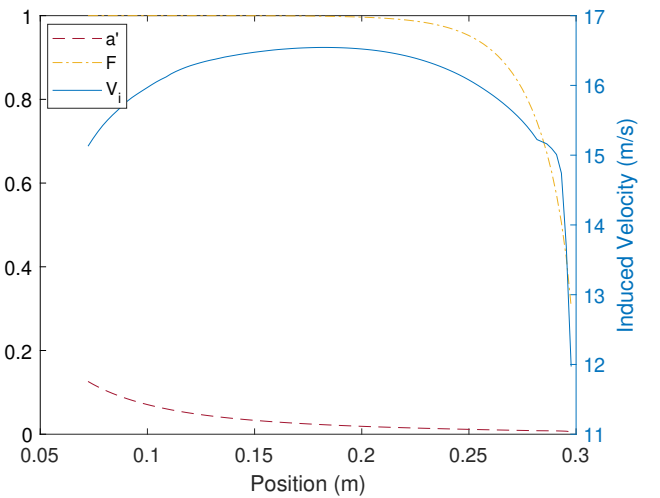


Figure 13: Variation of conditions along VTOL blade.

4.1 Pusher Thrust Lapse

To model the performance of the UAV it is necessary to determine the variation in maximum thrust available at different flight conditions. Firstly, Figure 14 shows the variation in the maximum thrust, as well as propulsive efficiency, as the freestream velocity increases. The maximum thrust capability and maximum efficiency occur close to the cruise velocity as designed. The propulsive efficiency during cruise is 0.89. This is likely an overestimation of the actual efficiency during operation due to the limitations of BEM theory as previously explained, as well as the blockage due to the tailplane and fuselage. Figure 14 also shows that the available thrust is always above 120 N above the stall speed of 29.7 m/s and therefore the thrust requirements for the transition and loiter phases are satisfied.

Figure 15 shows the variation of maximum thrust and efficiency at maximum thrust as altitude increases. This was considered with a fixed IC engine which decreases in power as given by Equation 9 [1], as well as the effects of changing density on the propellers. Using this data, the SEP plot could be produced which gave an absolute ceiling of 6.3 km for the drone.

$$P = P_0 \cdot \left(\frac{\rho}{\rho_0} - \frac{1 - \rho/\rho_0}{7.55} \right) \quad (9)$$

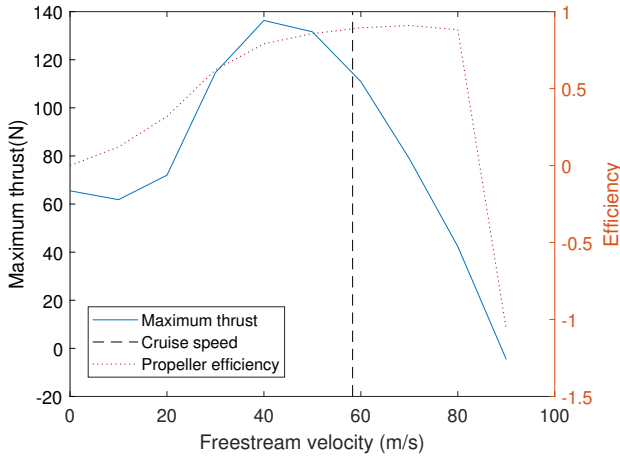


Figure 14: Variation of maximum thrust with freestream velocity for pusher propeller at $h = 0$.

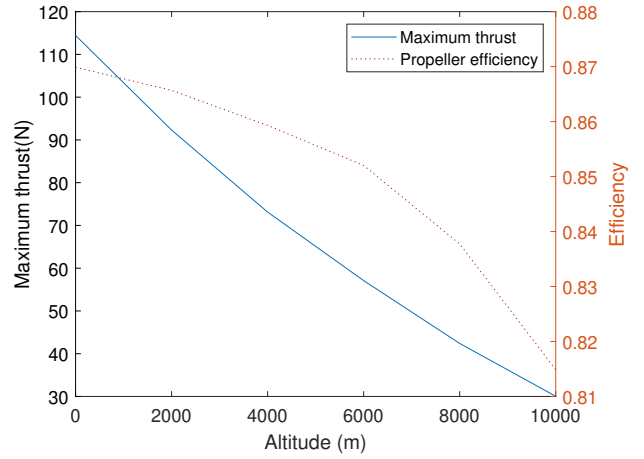


Figure 15: Variation of maximum thrust with altitude for pusher propeller at $V_\infty = 58.3$ m/s.

4.2 Effect of Forward Velocity on VTOL Propellers

The effect of forward velocity on the vertical propellers during the transition phase must also be modelled for aerodynamic and structural considerations. Therefore, the change in the thrust and torque produced by a single blade as the propeller performs a revolution are computed. To calculate this, the component of forward velocity perpendicular to the blade quarter-chord point is added to the perceived horizontal velocity. This changes the perceived velocity V , inflow angle ϕ , and local Reynolds number. It also increases the local Mach number which is one of the main reasons the limit RPM on the VTOL propellers is lower than on the pusher despite their equal radius as explained in the optimization process [6].

The oscillation in thrust and torque along a single blade is shown in Figure 16 and the added thrust and torque from both blades is shown in Figure 17. The highest forward velocity considered is 37 m/s. This was given by the mission requirements as it is the addition of stall speed and maximum expected gust speed. The worst-case structural scenario for the mission is for the propellers to be at maximum RPM, sustaining 100% of the weight, while at 37 m/s forward velocity. This is an overestimate of the normal operation. However, the wing may still be ineffective due to a gust stalling it and/or rotor downwash. Alternatively, the vertical rotor may not have had time to spool down. In any of these cases, the blades must be capable to withstand the loads. In this analysis it is especially important to take the limitations of blade element momentum theory into account when interpreting the results. Not only does BEM assume that the flow on each annulus of the rotor is independent, now it also assumes the flow on each angular position is independent. It also neglects the interaction between different blades, even if this should be minimized by having only two blades. Additionally, the component of forward velocity which is tangential to the blade, causing span-wise flow, is ignored by virtue of the assumption that flow at different blade sections is independent.

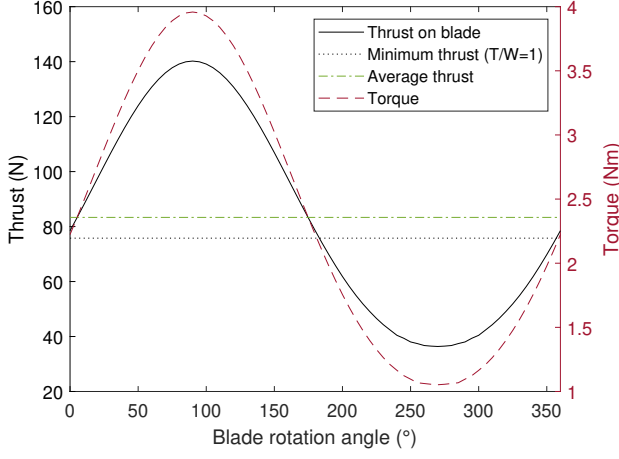


Figure 16: Variation of thrust on a VTOL blade as it spins at $V_{forward} = 37$ m/s.

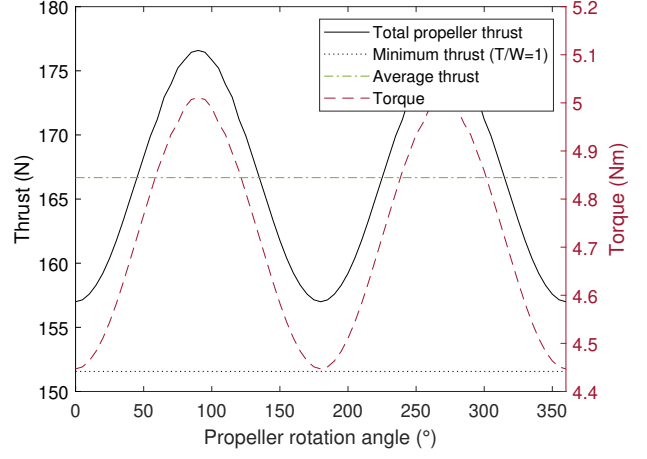


Figure 17: Variation of total thrust on a VTOL propeller as it spins at $V_{forward} = 37$ m/s.

The loads are highest when the blade is heading straight into the wind and these loads were considered to be the most constraining loads. Therefore, they were used as inputs for the structural optimization performed in [3] and for the FEA analysis. The case without forward velocity but at maximum rotation was also considered as it has the highest centripetal forces. It was seen, however, that at all blade sections the maximum forward velocity environment caused the highest loads.

The periodical variation in thrust induces fatigue loads. These are the most significant cyclical loads as the frequency is much higher than the other fatigue loads considered. Other cyclical loads are loads induced by wind gusts and each time the VTOL propellers spool up/down in a flight regime change. The frequency of the loads on the blade is 100 Hz as shown in Figure 16 while the frequency of the loads on the pylons, which is given by the total propeller loads, is of 200 Hz.

4.3 VTOL Thrust Lapse

The variation of the maximum thrust at different flight conditions was analyzed for the VTOL propellers similarly to the analysis performed for the pusher. The variation of maximum thrust with axial velocity is shown in Figure 18, where the limiting factor is the motor power, the analysis shows that the maximum climb velocity at MTOW ($T/W_0 = 1$) is 9.9 m/s. The variation of required power for hover as the forward speed increases was found to be negligible (0.13%) at the worst conditions and therefore is not plotted. However, the most constraining factor becomes the fixed RPM limit of 6000. Therefore, the variation of thrust and torque with forward speed at fixed maximum RPM is shown in Figure 19.

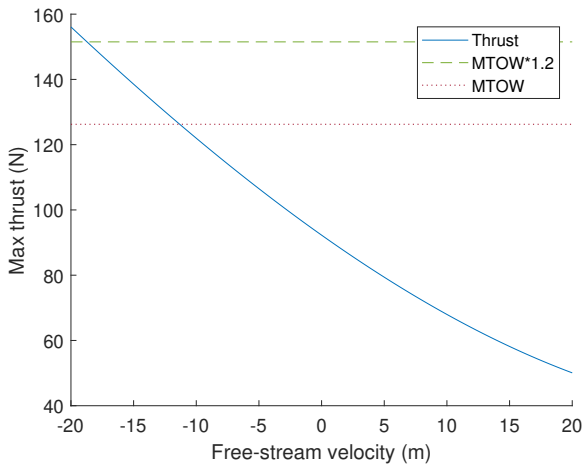


Figure 18: Variation of VTOL maximum thrust with changing V_{climb} at fixed power.

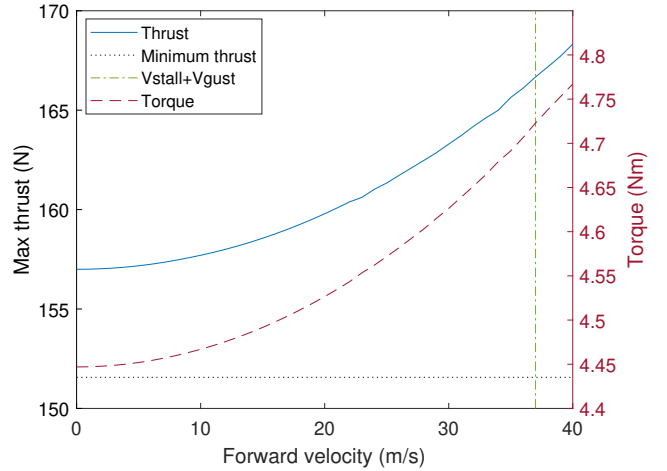


Figure 19: Variation of VTOL maximum thrust with changing $V_{forward}$ at fixed RPM.

The variation of thrust with altitude for the VTOL motors is not relevant for the mission profile as the VTOL motors only operate at very low altitude. However, it is provided in Figure 24 in Appendix A for completeness. As the VTOL propellers are powered by electric motors, the shaft power doesn't decrease with altitude and the only effect is the change in aerodynamic forces due to density.

4.4 VTOL OEI Condition

In the case where one of the electric motors fails, the performance has been analyzed to determine the type of response. The analysis was performed at the worst case scenario, when the drone is at MTOW and has no forward velocity and therefore the wings generate no lift. If the drone is at high altitude, one option is to transition to forward flight by using the available potential energy. If gravity alone is used, the minimum height to reach stall speed is 45.9 m as well as the height necessary to perform the rotation manoeuvre. This is the most effective option if the drone is at sufficiently high altitude. However, given mission profile, it is very unlikely to be applicable as the objective is to transition to forward flight as fast as possible to reduce battery requirements.

Another possibility is to use only the remaining three motors for vertical equilibrium. The thrust produced by the propellers is not sufficient during hover. A minimum descent rate of 5.57 m/s is required for three motors to produce a T/W ratio of 1. However, in this configuration there is a torque imbalance of 4.24 Nm. If the drone were to enter a spin, the drag in the vertical stabilizer would eventually counteract the torque. Modelling the vertical tailplane from the side as a flat plate, the steady rotation rate for equilibrium would be 5.85 rad/s. This is an overestimate as it neglects the drag on other components and the effect of yaw on the propellers. However, a spin of this magnitude would be uncontrollable due to the aerodynamic surfaces.

In a more realistic motor failure scenario, the propeller on the failed motor is still spinning due to inertia. A controller must be designed in a further design phase which would enter into forward flight and a controlled descent for autorotation on the unpowered propeller [7]. In autorotation, the change in angle of attack on the blades produces counteracting torque on the propellers which produces torque balance at sufficient speed. This would be similar to the autorotation manoeuvre used in helicopters. However, unlike a helicopter, there are still three active motors. Two of them still balance their torque and therefore the torque imbalance is lower. Additionally, thanks to the pusher, it is much easier to gain the forward speed necessary for autorotation.

Moreover, as forward speed increases, the aircraft is transitioning into its fixed-wing configuration with the wings producing lift and the control surfaces becoming effective. Current controllers developed by the Control subteam enter a descent soon after takeoff and after approximately 15 m/s of forward speed, the VTOL propellers start to spool down. In these conditions, the drone could maintain steady level flight with thrust only from three rotors. This speed is very quickly achieved with current maximum acceleration rates of 1.26 m/s produced only by the pusher. Adding the speed gained due to the controlled descent means the aircraft will achieve a sustainable steady level flight in less than 12 seconds.

Therefore, this means that the drone is able to perform a controlled transition into forward flight without losing a large amount of altitude and therefore this manoeuvre can be performed safely except at extremely low altitudes. Oppositely, if a failure occurs at very low altitude during takeoff or landing, the drone is able to perform a controlled emergency landing before the propeller loses significant rotational speed and is still controllable only with the remaining VTOL motors.

Overall, given the quadcopter configuration, the pusher propeller and the aerodynamic surfaces, the drone will be able to recover from a loss of power on a single motor at any flight stage with appropriate design.

5 Finite Element Analysis

An FE analysis of the propeller blades was performed at the maximum loading conditions. This analysis serves to validate the results obtained in the analytical structural analysis and optimization explained in [3]. The results obtained analytically should be accurate for the stresses due to bending and centrifugal force. However, as the cross-section of the blade is not thin-walled, it's not possible to obtain results for the stresses due to torsion. This effect can be especially relevant given the highly cambered airfoil chosen for the blades. Therefore the FE simulation is necessary to ensure structural integrity.

Firstly, CADs of the pusher and VTOL blades were created in SolidWorks [14] as guided lofts as shown in Figures 5 and 6. Apart from providing a shape for the simulation, SolidWorks also calculates the mass properties of the blade. Each pusher blade has a mass of 51.7 g and a moment of inertia of $2.72 \cdot 10^{-4}$ kg·m². Each VTOL blade has a mass of 217 g and a moment of inertia of $7.85 \cdot 10^{-4}$ kg·m². These values as well as centroid data were then used to perform spool-up time estimations and weight and balance as shown in [4]. The material selection is also shown in [4].

Abaqus [15] simulations were then set up with the models. The loads calculated with blade element momentum theory in the most constraining scenarios were applied in the simulation. The loads were applied to

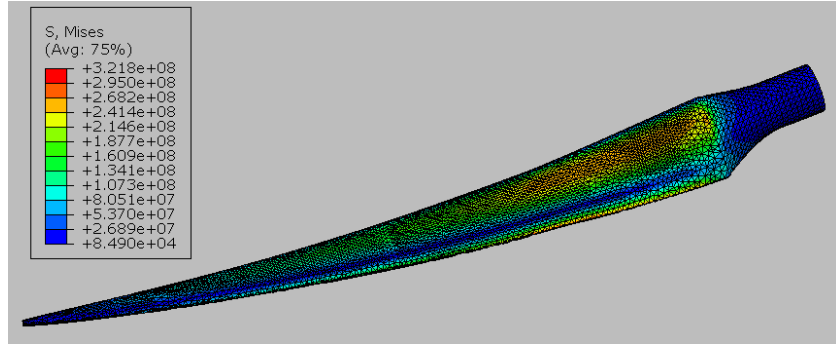


Figure 20: Von Mises stress on pusher blade in FEA.

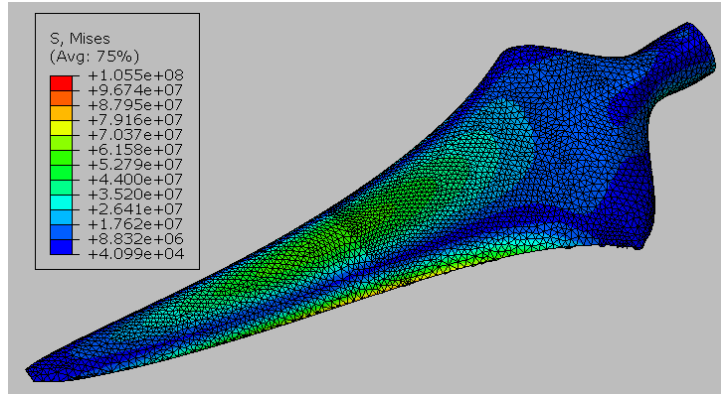


Figure 21: Von Mises stress on VTOL blade in FEA.

several reference points distributed along the quarter-chord line of the blade. The reference points were tied to the body with a structurally distributed coupling. The loads applied at each reference point contained the lift, drag and torque generated by each blade section. Additionally, centrifugal forces were applied at the maximum rotation rate. A simple mesh convergence study was performed to ensure converged results.

The stresses in the blades are shown in the deformed plots in Figures 20 and 21. It can be seen that the maximum stress in the pusher blade is 322 MPa. Even with a 1.5 safety factor the maximum stress is below the minimum yield strength of Aluminium 7068 of 648 MPa. The maximum stress in the VTOL blade is 105 MPa, well below the limits for the material. This was expected as the blade has a much thicker root from the aerodynamic optimization and didn't require any structural modifications.

The importance of the FEA analysis can be seen in Figure 20. There are two main high-stress regions. Firstly, the top of the blade near the root (and the bottom) can be seen to be very stressed. This was predicted by the analytical structural calculations due to vertical bending and the appropriate chord adjustments were made to sustain this stress. However, there is a second high-stress area in the trailing edge near the root which turns out to be the most stressed area in the structure. This stress is not caused by horizontal bending as the moment of inertia of the blade along in the chord axis is much larger than along the thickness. Therefore simple beam theory predicted high stress on the top and bottom of the blades but not in the trailing and leading edge. The stress appears however due to the torsion caused by the high pitching moments along the blade. The stress couldn't be predicted analytically but the FEA confirms that the structure can withstand the loads including torsional effects.

The next phenomenon which was analyzed was the change in angle of attack due to structural deformation. There are two causes for this change in angle. Firstly, the negative pitching moment directly causes torsion which reduces the angle of attack in the outer sections of the blade. The second cause is that lift and drag act through the quarter-chord line which is significantly aft of the shear centre thereby causing a positive moment which increases the angle of attack.

For the VTOL propeller, at half the span of the blade, there is an increase in twist angle of 0.19° due to the aerodynamic loads. At the tip there is a change of -0.15° . This is consistent with the fact that closer to the root the bending moments are more significant and therefore the change in angle should be net positive. Oppositely, near the tip of the blade the cross-section becomes very thin making torsion the dominant phenomenon. Overall, the magnitude of the angle changes is very low and should not cause significant aeroelastic effects.

On the other hand, for the pusher blade, the angle changes are 2.3° and 1.01° at the middle of the blade and at the tip respectively. This is because the pusher blade is much thinner making bending more significant throughout the whole blade. This deformation is much more significant than for the VTOL propellers as a change in angle of attack of 2.3° significantly affect the lift to drag ratio. Therefore, in further iterations of the design, it will be necessary to either strengthen the structure to reduce this effect or adjust the twist to counteract the aerodynamic deformation to achieve the desired flow conditions.

6 Control Surface Actuators

The second aircraft component covered in this report is the selection and integration of the flight control surface actuators. There are broadly two categories of actuators for aerospace applications, electric and hydraulic. For relatively small UAVs it is impractical to have a hydraulic system due to the increased weight and maintenance required. Additionally, as the system is already hybrid-electric for the VTOL motors, large batteries are already available to supply the actuators. Therefore, electromechanical servos were selected [16].

The control surface geometries and maximum hinge moments are given by control and stability requirements and are shown in Table 1. The actuators must be selected and integrated such as to satisfy the moment requirements and size limitations. Within the geometric constraints, the main objective is to minimize the mass of the actuators as weight is very penalizing for this mission but cost is also considered.

Within electric actuators there exist linear actuators which produce a force and angular actuators which produce a moment. Angular servos are usually more appropriate for small drones as they can be attached to the structure and don't require a tilting support like linear actuators.

The transmission of the loads from the actuator to the control surface is also important to consider. Initially, connecting the servos directly to the hinges was attempted. However, the estimated maximum torques increased greatly as better stability requirements were calculate from weight and balance estimations. Therefore, a transmission system with a pushrod was selected. This allows the actuators to be placed between the two spars of the aerodynamic surface and only requires a small cutout in the rear spar for the pushrod.

The pushrod is attached to the servo via the servo arm and to the hinge via the control horn. The sizing of these components has significant consequences. The required moment of the actuator is given by $T_{act} = L_{horn}/L_{arm} \cdot T_{hinge}$, therefore, increasing the L_{horn}/L_{arm} ratio decreases the actuator requirements. However, it also decreases the angular displacement of the hinge as the servo spins. For this reason a ratio of $1/2$ was selected for both the aileron and elevator actuators. For the rudder, a ratio of $1/3$ was chosen instead. This was decided as the available thickness in the vertical tail is only 1.2 mm making reducing the required torque especially important. This choice means the maximum available deflection at the rudder is approximately 25° , significantly less than the 40° available at the other control surfaces. However, it is more deflection than the required 13.8° for controllability in the most demanding scenario.

Additionally, if the hinge torque is too high to be supplied by a single actuator for the given size requirements, there are two options. More than one actuator could be added to a control surface or a control surface could be split into different sections with different actuators. This reduces the required torque per servo but increases the amount of transmission structures, controllers, and cabling needed.

Considering the previous trade-off, it was chosen to have two different actuators for each aileron as they have a very high moment requirement. One is situated at the beginning of the aileron and the other is at the end as shown in Figure 23. Moving weight further outboard on the wings doesn't increase the structural requirements of the wings as they generate lift along the span. Furthermore, cabling is already present at that span as the pylons for the VTOL motors are further outboard than the ailerons.

Oppositely, for the two elevators and two rudders, it was chosen to have only one actuator for each control surface placed as close to the root as possible. This was decided because the H-tail empennage is very lightweight and having actuators further outward significantly increases the structural requirements. Additionally, both surfaces are tapered which makes placing actuators further outboard more geomtrically constrained.

The sizing of the pushrod is also significant, as increasing its radius reduces the maximum control horn length. Aluminium 2024 was selected for the moment transmission structure due to its high E/ρ merit index and its high corrosion resistance. The expected failure mode for the rods is buckling as they are slender and are subjected to high compressive loads. Equation 10 shows the buckling load for a uniform cross-section beam with pinned boundary conditions as well as the second second moment of area of a solid cylinder. Using these relations the minimum required diameter, D , for the pushrods was calculated taking into account a 1.5 ultimate load safety factor. The rods can also be in tension as the surfaces deflect in the other direction, in this case, the stresses in the rods are well below the yield stress as the buckling requirement is much more demanding.

$$P_{crit} = \frac{\pi^2 EI}{L^2} \text{ where } I = \frac{\pi D^4}{64} \quad (10)$$

Figures 22 and 23 show the side and top view of the aileron actuator setup. Similar figures for the elevator and rudder are provided in Appendix B. As it can be seen, the control horns were sized to not exceed the thickness of the control surfaces. Therefore, the actuator systems are fully contained within the aerodynamic surfaces and no aerodynamic fairings must be added to accommodate it. The actuators were placed as close as possible to the rear spar to reduce the pushrod length and placed as low as possible (within skin thickness requirements) to reduce the pushrod angle.

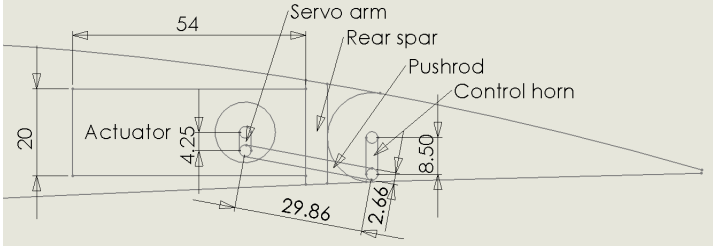


Figure 22: Side view of elevator actuator system. All sizes in mm.

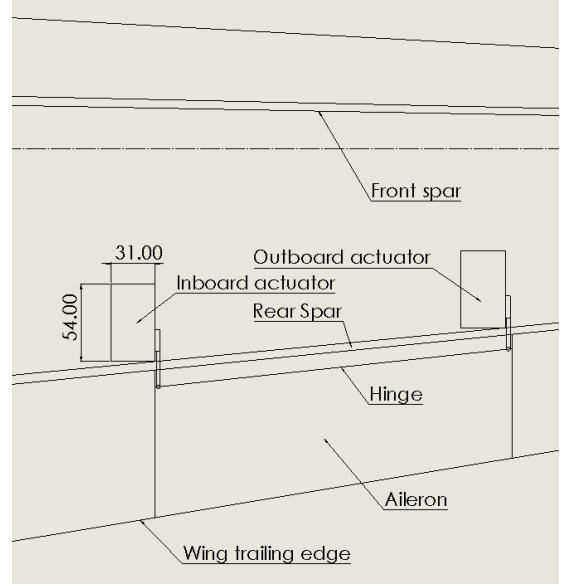


Figure 23: Top view of elevator actuator system. All sizes in mm.

Table 1: Actuator properties.

	Ailerons [19]	Elevators [18]	Rudders [17]
Maximum hinge moment (Nm)	13.8	4.2	1.37
Chosen servo	BLS04L	Hyperion DS20	Graupner DES 448
Number of control surfaces	2	2	2
Number of servos	4	2	2
L_{arm}/L_{horn}	0.5	0.5	0.333
Required torque per servo (N)	3.45	2.1	0.456
Torque per servo (N)	3.77	2.3	0.46
Voltage (V)	8	6	6
Pushrod length (mm)	29.9	26.6	11.3
Pushrod radius (mm)	2.7	2.1	2
Servo size (mm)	40 x 20 x 27	40 x 19.6 x 38	31.5 x 9 x 33
Weight per servo (g)	65	53	19
Maximum control surface deflection ($^{\circ}$):	40	40	25
Cost per servo (£)	76.05	31.2	48.9
Pushrod weight (g)	7.42	1.99	0.77

Table 1 shows the selected actuators as well as their properties. The total weight of the servos is 404 g, the total weight of the carrying structures is 10.2 g and the total cost of the servos is 465 £. Initially, it was attempted to have a uniform voltage across actuators to require only one step-down transformer. However, due to the large difference in required moments, this was not possible. Nonetheless, only two separate systems are needed, one 8 V circuit for the aileron actuators and one 6 V circuit for the actuators in the empennage.

The last consideration to be made about actuators is their response time. All three selected actuator types have response times below 0.5 s for a 60° rotation which is greatly in excess of all requirements.

7 Discussion and Further Steps

The design process for this project began with the mission profile definition and baseline configuration selection. After selecting the baseline configuration, the initial sizing was performed to estimate the drone's size and requirements. However, after this design iteration, there are significant considerations to be made:

- Firstly, the weight estimation of the aircraft differed slightly from the predicted amount. This is mainly caused by two factors. The final battery fraction is much smaller than expected, a 21% battery fraction was initially expected but the final value was found to be 10.5% as shown in [3]. This caused a very significant overestimation of the total weight. However, the accuracy of the regression data used to estimate the empty weight fraction must also be scrutinized. Most small UAVs have either only a quadcopter or a fixed-wing configuration. Therefore, it was to be expected that the weight fractions of systems and load-bearing structures would be higher than previous UAVs. Overall, mainly due to the compound effect of these two factors, a total weight of 49 kg was obtained in weight and balance [4]. This value is 2.5 kg less than the initial estimate implying that the drone is overdesigned in terms of aerodynamic surfaces and structures. In a further design iteration, with accurate weight fractions from this design iteration's weight and balance, the weight could be decreased significantly.
- Secondly, the magnitude of C_{d_0} was underestimated. This was also caused by the scarcity of regression data for aerodynamic properties of fixed-wing VTOL drones. Additionally, the fuselage is very wide to accommodate the large payload. In a further design iteration, it would be appropriate to consider the C_{d_0} obtained by CFD simulations performed by the aerodynamics team. Together with the new weight from the previous consideration, an updated thrust requirement can be obtained and the propellers and engine resized accordingly.

The following stage in the report was propeller analysis and refinement. Overall, the design satisfies all the requirements with good performance metrics. However, there are several improvements to be made both in the analysis and the design:

- The majority of the analysis was based on blade element momentum theory, this is very effective for initial design and analysis. However, it must be complemented with CFD analysis of the blades in further design process. This would capture many effects which BEM neglects. For example, the interaction between the rotors and the motors/pylons or the fuselage/tailplane, the effects of compressibility as the Mach number is over 0.3 on much of the blade, and it could better capture the unsteady oscillations in thrust in the VTOL rotors during forward flight.
- The optimization process for the blade design also had significant limitations. In essence, it consisted of maximizing aerodynamic efficiency within geometric constraints and then readjusting the chord if it was necessary for structural integrity. This approach, although effective at minimizing power requirements, neglects other aspects of design which hindered many properties of the VTOL propellers in particular.
- A spool-up time of approximately 20 s for the VTOL motors and a mass of 217 g per blade was obtained from inertial values. The long spool-up time creates significant controllability issues. The four VTOL motors use variable RPM to control pitch, roll, and yaw during the VTOL phase. The very slow response time poses significant difficulty for effective reaction to perturbations. Different options exist to address this issue. Having variable pitch rotors would greatly increase control but would create the need for many additional systems, especially if the four propellers have this capability. Alternatively, if variable RPM control is maintained, minimizing the moment of inertia of the propellers should become an important design driver and must be balanced with aerodynamic performance.
- Similarly to the previous consideration, a mass of 217 g per blade creates a significant weight penalty for the aircraft. The weight is located mostly near the root of the blade where very few lift is generated. Additionally, the real thrust in this area will be even less than BEM suggests due to blockage from motors. Therefore, having such a large chord, way above the structural requirements, just to satisfy the optimal aerodynamic distribution, is not the optimal decision in this context. A small decrease in aerodynamic performance accompanied by a large decrease in mass is favourable for the aircraft in terms of total power required.
- Overall, a more appropriate blade optimization process should strive to maximize aerodynamic efficiency balanced with reducing mass for the overall performance of the aircraft. It must also aim to reduce the inertia of the motors for better control capabilities, which in turn will decrease power requirements.

Finally, the last component covered in this report was the sizing and integration of the actuators. Although the mass and cost of the actuators was satisfactorily low, further optimizations can be made, especially for the ailerons. In the current design they are placed inboard of the VTOL pylons in the wing, creating a very large moment requirement in comparison with similar aircraft. Relocation of this control surface further outboard in a further design iteration would significantly increase the efficiency of the surfaces for roll control, reducing the surface size and the actuator requirements.

In conclusion, the propulsion of the aircraft meets all the thrust requirements to complete the mission with very competitive power requirements. However, further iterations of the design should consider more accurate data for initial sizing and design a propulsive system which achieves a better balance between propulsive efficiency, mass, and controllability.

References

- [1] Raymer D. Aircraft Design: A Conceptual Approach, Sixth Edition. Washington, DC: American Institute of Aeronautics and Astronautics, Inc; 2018.
- [2] Roskam J. Airplane design. 3. print., rev. November 4, 2002 ed. Lawrence, Kansas: DARcorporation; 2000.
- [3] Fernández N. Propulsion and systems. Technical report, Imperial College London, 2022.
- [4] Lee W. Propulsion and systems. Technical report, Imperial College London, 2022.
- [5] Rieuneau A. Structures. Technical report, Imperial College London, 2022.
- [6] Hamburger M. Propulsion and systems. Technical report, Imperial College London, 2022.
- [7] Johnson W. Rotorcraft aeromechanics. Cambridge aerospace series Cambridge [u.a.]: Cambridge University Press; 2013.
- [8] Jamieson P. Innovation in wind turbine design. 1. publ. ed. Chichester: Wiley; 2011.
- [9] Branlard E. Tip-Losses with Focus on Prandtl's Tip Loss Factor. Wind Turbine Aerodynamics and Vorticity-Based Methods. Cham: Springer International Publishing; 2017. pp. 227-245.
- [10] Adkins CN, Liebeck RH. Design of optimum propellers. Journal of propulsion and power. 1994; 10 (5): 676-682. Available from: doi: 10.2514/3.23779 Available from: <http://arc.aiaa.org/doi/full/10.2514/3.23779>
- [11] Drela M, MIT. XFOIL (6.99) 2013.
- [12] Erturk E. A Spreadsheet Tool for the AERODAS Model for Calculating Airfoil Pre-Stall and Post-Stall Lift and Drag Characteristics . INTERNATIONAL JOURNAL of RENEWABLE ENERGY RESEARCH. 2018; 8 (4)
- [13] Spera DA. Models of Lift and Drag Coefficients of Stalled and Unstalled Airfoils in Wind Turbines and Wind Tunnels. 2008.
- [14] Dassault Systèmes Simulia Corp. SolidWorks. 2021.
- [15] Dassault Systèmes Simulia Corp. Abaqus. 2022.
- [16] Gundlach J. Designing Unmanned Aircraft Systems: A Comprehensive Approach, Second Edition. Washington, DC: American Institute of Aeronautics and Astronautics, Inc; 2014.
- [17] Servodatabase. Graupner DES 448. Available from: <https://servodatabase.com/servo/graupner/des-448>.
- [18] Servodatabase. Hyperion DS20-UMD. Available from: <https://servodatabase.com/servo/hyperion/ds20-umd>.
- [19] KINGMAX HOBBY Co. L. BLS04L MANUAL. Available from: <https://www.unilight.at/images/KINGMAX/BLS04LMANUAL.pdf>.

Appendix

A VTOL Thrust Lapse With Altitude

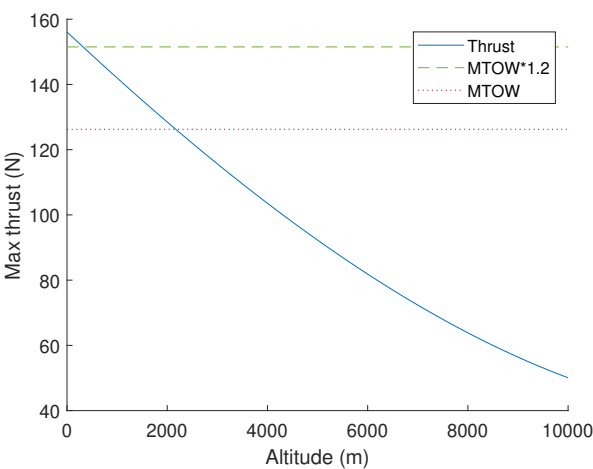


Figure 24: Variation of VTOL maximum thrust with altitude.

B Actuator Drawings

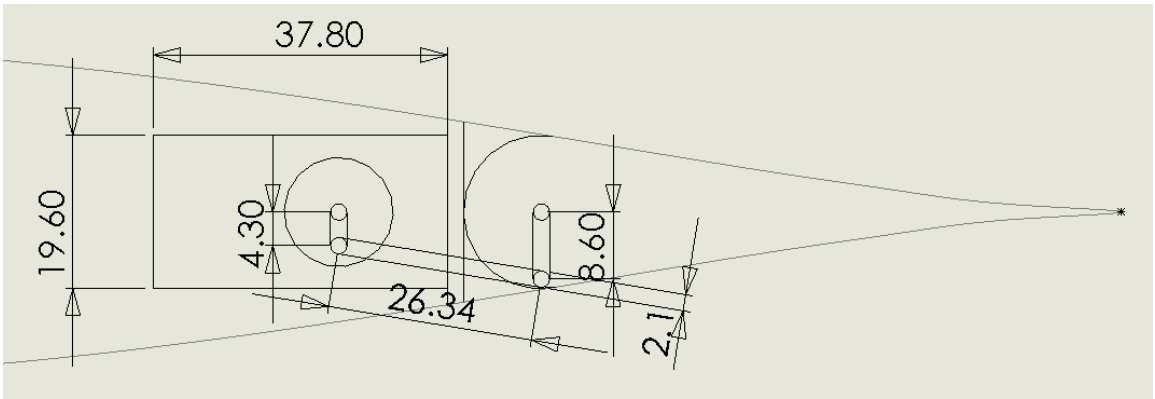


Figure 25: Side view of elevator actuator. All sizes in mm.

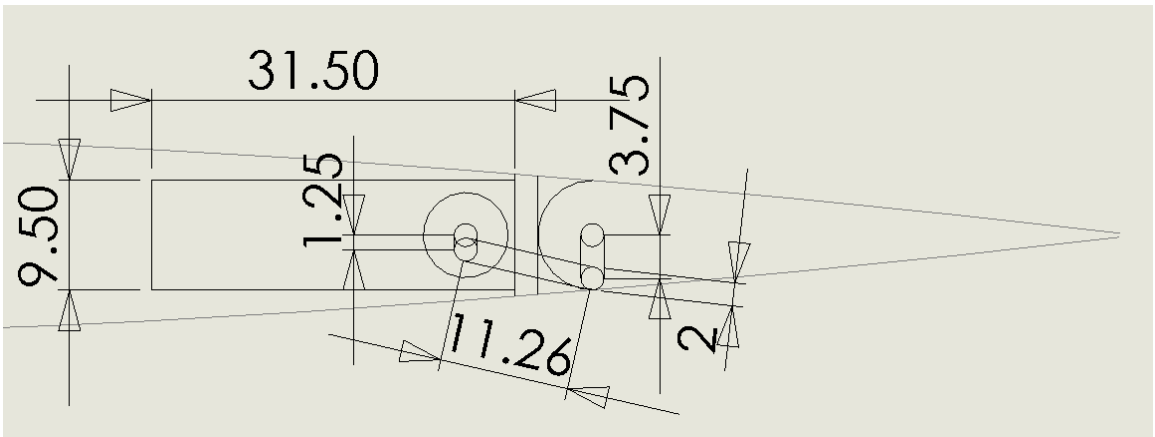


Figure 26: Side view of rudder actuator. All sizes in mm.